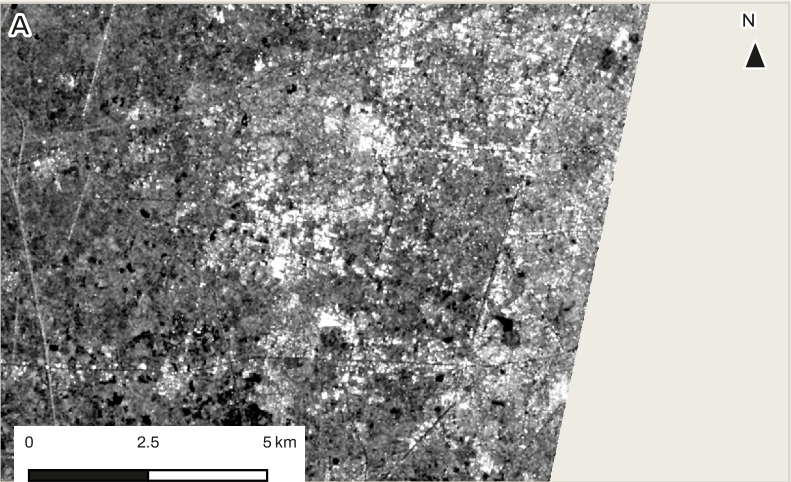


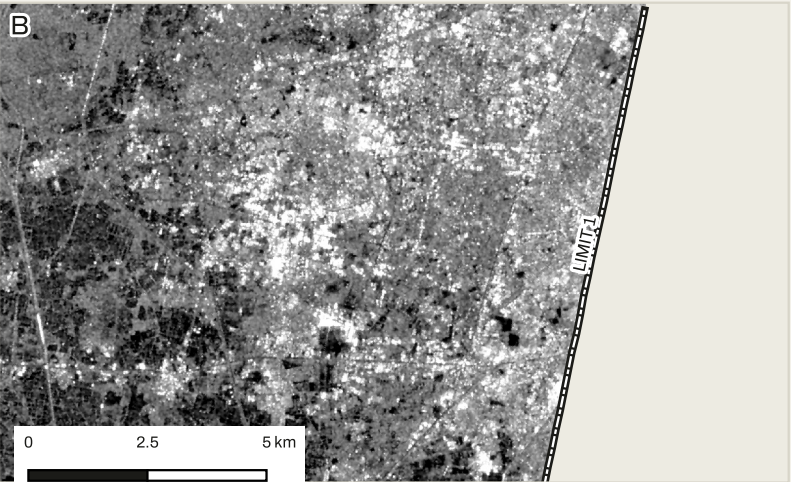
On 25 July the radar's new open water lay in farmland; the flooded built-up belt was hidden from it

Western Ahmedabad (Bopal, Ghuma, Shela, Sarkhej). Radar VV before and during the flood, same orbit and geometry; change; and the classified result.

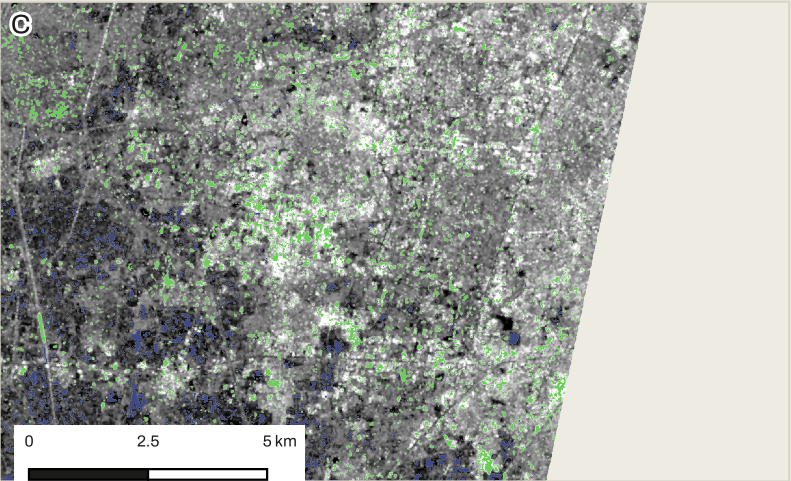
Prepared by Sahasrik Ragani



A · Radar VV, 13 Jul (before)
Sentinel-1D C-SAR IW GRDH · VV gamma0 dB · 2026-07-13 01:09Z · 10 m · S1D_IW_GRDH_1SDV_20260713T010924_20260713T010949_003648_006824_7405



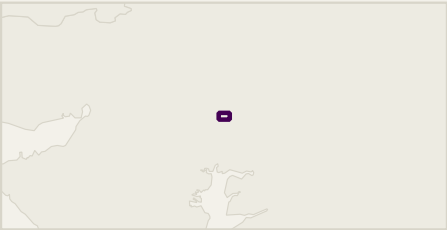
B · Radar VV, 25 Jul 06:39 IST, in light rain
Sentinel-1D C-SAR IW GRDH · VV gamma0 dB · 2026-07-25 01:09Z · 10 m · S1D_IW_GRDH_1SDV_20260725T010925_20260725T010950_003823_006E26_FBB7



C · VV change from the 1 and 13 Jul mean, beyond 4.8 dB
Sentinel-1D C-SAR IW GRDH · VV change vs mean of 1 and 13 Jul, dB · 2026-07-25 01:09Z vs 2026-07-01 and 2026-07-13 01:09Z · 10 m · S1D_IW_GRDH_1SDV_20260725T010925_20260725T010950_003823_006E26_FBB7

Sources

A Sentinel-1D C-SAR IW GRDH · VV gamma0 dB · 2026-07-13 01:09Z · 10 m · S1D_IW_GRDH_1SDV_20260713T010924_20260713T010949_003648_006824_7405 — sarsen 0.9.6 terrain correction to gamma0 on GLO-30 (geoid-corrected to ellipsoid), 10 m UTM 43N; Lee 5x5 filter
B Sentinel-1D C-SAR IW GRDH · VV gamma0 dB · 2026-07-25 01:09Z · 10 m · S1D_IW_GRDH_1SDV_20260725T010925_20260725T010950_003823_006E26_FBB7 — sarsen 0.9.6 terrain correction to gamma0 on GLO-30 (geoid-corrected to ellipsoid), 10 m UTM 43N; Lee 5x5 filter
C Sentinel-1D C-SAR IW GRDH · VV change vs mean of 1 and 13 Jul, dB · 2026-07-25 01:09Z vs 2026-07-01 and 2026-07-13 01:09Z · 10 m · S1D_IW_GRDH_1SDV_20260725T010925_20260725T010950_003823_006E26_FBB7 — sarsen 0.9.6 terrain correction to gamma0 on GLO-30 (geoid-corrected to ellipsoid), 10 m UTM 43N; Lee 5x5 filter
D Sentinel-1D C-SAR IW GRDH · classified: new open water / built-up brightening · 2026-07-25 01:09Z · 10 m · S1D_IW_GRDH_1SDV_20260725T010925_20260725T010950_003823_006E26_FBB7 — sarsen 0.9.6 terrain correction to gamma0 on GLO-30 (geoid-corrected to ellipsoid), 10 m UTM 43N; Lee 5x5 filter; classes from studies/ahmedabad/sar.py



Gujarat, India; box = panel A

How to read the marks

- OBS**
measured in these pixels; sensor, date and GSD cited
- INF**
what the observations imply; reasoning in the evidence field
- CLAIM**
the object of study, not evidence
- LIMIT**
footprint or positional uncertainty of a measurement

Marks

LIMIT 1 Edge of the radar swath: nothing east of this line was imaged
OBS 2 New open water on 14% of the judgeable ground in this 1 km square; 100% of that water is on cropland (Sentinel-1D C-SAR IW GRDH, 2026-07-25 01:09Z, 10 m)
CLAIM 3 Claimed: Bopal-Ghuma-Shela worst hit, water about 3 ft deep for about 3 days (DeshGujarat 25 Jul 2026 (AMC); Counterview 5 Aug 2026 (opinion column))
INF 4 Built-up ground here brightened 0.4–1.0 dB more than at Thaltej, Bodakdev and Gota (from: median built-up change within 1.5 km (sar_e1_places.csv). Water at wall bases (double bounce) or building type; experimental.)

C: VV fell (dark blue) or rose (green) by 4.8 dB or more; Otsu threshold over the 67 most-darkened 640 m tiles. D, dark blue: new open water (also 25 Jul VV at most -12.8 dB; built-up, trees, permanent water excluded). D, green: built-up that brightened 3 dB or more (chosen; experimental).

What these images cannot show

Water between buildings: built-up land (WorldCover 2021) is excluded, so Bopal's streets are not scored. Rain at the pass roughens water and can hide it. Flood water in farmland and paddy flooded for transplanting look alike; 50% was still dark on 6 Aug. The swath misses the river and the east. 10 m pixels miss underpasses.